Secondary polytopes and othertopes

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Josef Ehm, *The Terrace Café at Barrandov* (1936)

What is the moduli space of all triangulations of a polytope?

*In modern terms, a **moduli problem** consists of a class of algebraic objects—schemes, subschemes of a given scheme, maps of schemes, sheaves on schemes, typically defined by some common attibutes—and a notion of what it means to have a **family** of these objects parameterised by a scheme B…*

*Typically what we want from a solution to a moduli problem is to understand all possible families of the objects in question. In particular, the individual objects should be in natural one-to-one correspondence with the closed points of the moduli space. The word **natural** is the key.*

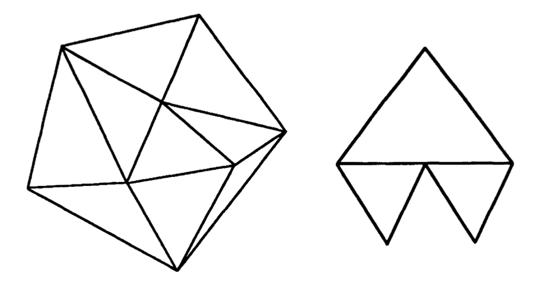
Eisenbud & Harris, The practice of algebraic curves, AMS (2024), §§7.1–7.2

Definition

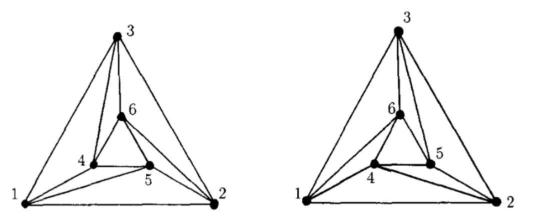
$$\mathcal{A} = \{a_1, ..., a_n\} \subset \mathbb{R}^d.$$

A **triangulation** of $\mathcal{A}$ is a triangulation of the convex hull conv($\mathcal{A}$) with vertices in $\mathcal{A}$.

Given a fixed triangulation Δ of conv $\mathcal{A}$, any $\psi \in \mathbb{R}^n = \mathbb{R}^{\mathcal{A}}$ defines a unique piecewise linear function $g_{\psi,\Delta}$: conv $\mathcal{A} \to \mathbb{R}$. Its graph is given by draping a sheet on the columns of height $\psi(a_j)$ above the point $a_j in \mathbb{R}^d$.
Δ is called **regular** if it is the projection of the graph of a *concave* piecewise linear function conv $\mathcal{A} \to \mathbb{R}$.



[GKZ, fig. 22]



[BFS, fig. 1]

Definition

The **characteristic function** of Δ is the function $\psi_\Delta : \mathcal{A} \to \mathbb{R}$ defined by

$$\psi_\Delta(\omega) = \sum_{\sigma : \omega \in V(\sigma)} \mathrm{Vol}(\sigma)$$

(summation over all maximal simplices for which ω is a vertex).
The **secondary polytope** $\Sigma(\mathcal{A})$ is the convex hull in $\mathbb{R}^n = \mathbb{R}^{\mathcal{A}}$ of all characteristic functions.

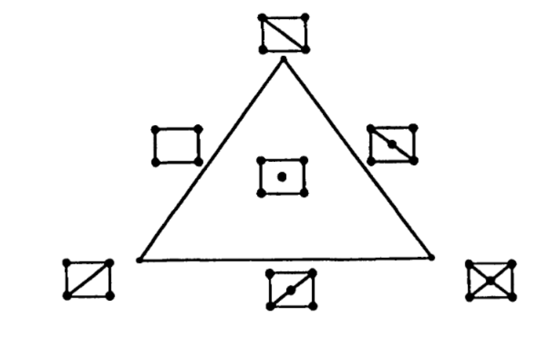
'Moduli space' of triangulations of $\mathcal{A}$.

Theorem (Gel'fand–Kaparanov–Zelevinsky, '90)

- *The secondary polytope has dimension $n - d - 1$.*
- *The vertices of $\Sigma(\mathcal{A})$ are exactly the characteristic functions of regular triangulations. Faces correspond to coherent subdivisions.*
- *For any triangulation Δ, the normal cone to $\Sigma(\mathcal{A})$ at ψ_Δ is the set of functions $\psi : \mathcal{A} \to \mathbb{R}$ such that*
  1. *$g_{\psi,\Delta}$ is concave*
  2. *for any $\omega \in \mathcal{A}$ which is not a vertex of a simplex in Δ, $g_{\psi,\Delta}(\omega) \geq \psi(\omega)$.*

We neglect the proof, but it's not too hard.

Five points in $\mathbb{R}^2$: $n - d - 1 = 5 - 2 - 1 = 2$.



[GKZ, fig. 28]

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Theorem (Billera–Sturmfels, '92)

Let T be the standard $(n-1)$-simplex in $\mathbb{R}^{\mathcal{A}}$, i.e. $T = \operatorname{conv}\{e_{a_1}, \dots, e_{a_n}\}$. Let $\pi : T \to \operatorname{conv} \mathcal{A}$ be the affine projection

$$\pi(e_{a_i}) = a_i.$$

Then

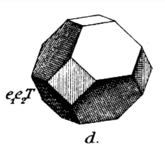
$$\begin{aligned}\Sigma(\mathcal{A}) &= (d+1) \int_{\operatorname{conv}\mathcal{A}} \pi^{-1}(x)\, dx \\ &:= (d+1)\left\{ \int_{\operatorname{conv}\mathcal{A}} \gamma(x)\, dx : \gamma \text{ is a section of } \pi \right\}\end{aligned}$$

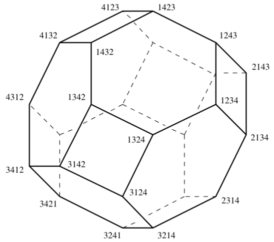
The secondary polytope is the **fibre polytope** of $T \to \operatorname{conv}\mathcal{A}$. Each POINT of a fibre polytope comes from a SECTION of $P \to Q$.

The permutahedron:
Convex hull in $\mathbb{R}^d$ of all vectors coming from permuting the coordinates of

$$(1, 2, \dots, d)^t$$

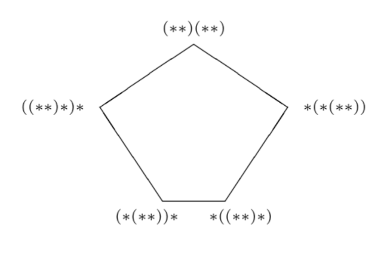
Vertices = permutations in S_d.
Two vertices joined by an edge $\iff$ they differ by an adjacent transposition.





L: Schoute (1911) repro. in [Z, p.17]. R: [Z, p.18]

The associahedron:
Vertices = different ways of bracketing a string of n letters = words of n letters if multiplication is not associative.
Two vertices joined by an edge $\iff$ they correspond to one application of the associative law.



[Z, p.18]

Theorem (Gel'fand–Zelevinsky–Kapranov, '90)

An associahedron is the secondary polytope of the cyclic n-gon:

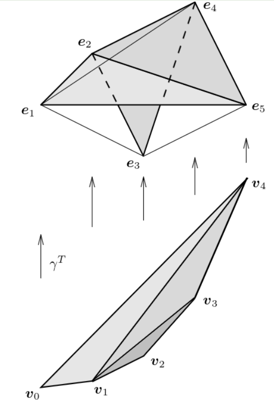


fig. from [Z] p.307

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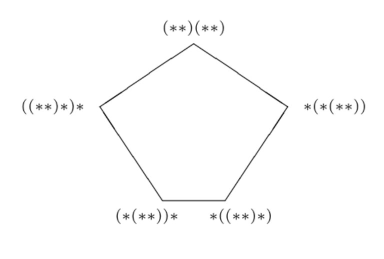

[Z, p.18]

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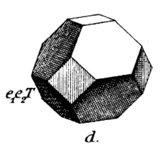

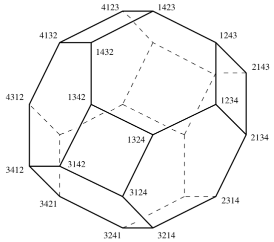

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Theorem (Billera-Sturmfels, '92)

A permutahedron is the fibre polytope of the projection of a cube onto an interval:

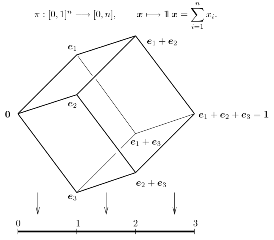

fig. from [Z] p.301

$$Permuta(1, \dots, n) = \int_{[0,1]} \pi^{-1}(x)\, dx$$

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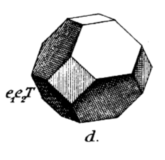

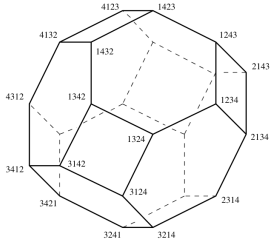

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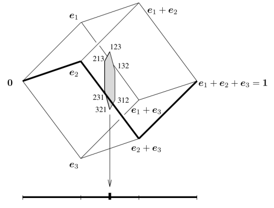

fig. from [Z] p.304

$$Permuta(1, \dots, n) = \int_{[0,1]} \pi^{-1}(x)\, dx$$

Vertices = increasing paths in the 1-skeleton of the cube = 'extremal' sections of the projection.

Bedtime reading

GKZ I. M. Gelfand, M. M. Kaparanov, and A. V. Zelevinsky. *Discriminants, resultants, and multidimensional determinants*, Chapter 7. Birkhaüser, 1993.

Z Günter M. Ziegler. *Lectures on polytopes* (7th pr.), Chapter 9. Grad. Texts. Math. 152. Springer, 2007.

BFS Louis J. Billera, Paul Filliman, and Bernd Sturmfels. “Constructions and complexity of secondary polytopes”. *Adv. Math.* **83**, 155–179 (1990).

BS Louis J. Billera and Bernd Sturmfels. “Fiber polytopes”. *Ann. Math. (2)* **135**, 527–549 (1992).